西湖大學 WESTLAKE UNIVERSITY

Bioinformatics training — 4

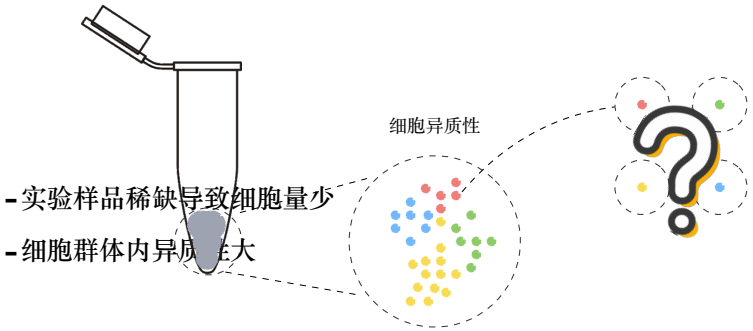
RNA-seq数据分析

Chengchen Zhao
Dec 30th, 2021

单细胞RNA-seq数据分析

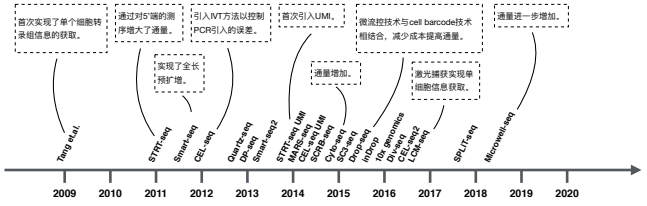
- 单细胞RNA-seq发展历史
- Dr.seq2软件介绍
- 利用Dr.seq2分析单细胞 RNA-seq 数据
- Dr.seq2参数说明
- Dr.seq2结果展示

为什么要做到单细胞尺度?

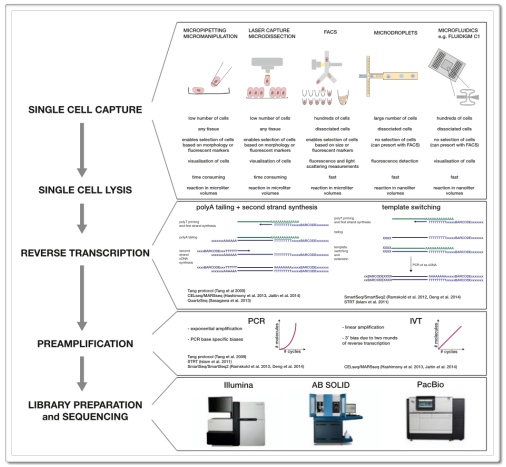


- 实验样品稀缺导致细胞量少
- 细胞群体内异质性大

单细胞转录组测序技术的发展

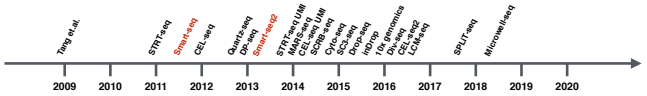


单细胞RNA-seq通用流程

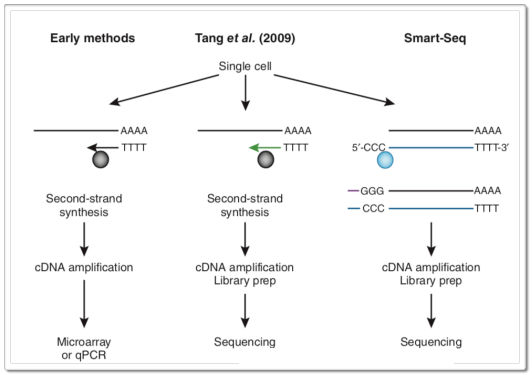


- 首先，进行单个细胞的捕获或分离；
- 然后进行细胞溶解；
- 随后对细胞中的RNA进行反转录；
- 之后对反转录形成的cDNA进行预扩增；
- 最后完成文库构建并进行测序。

单细胞RNA-seq技术举例—Smart-seq

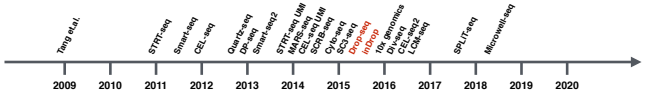


单细胞RNA-seq技术举例—Smart-seq

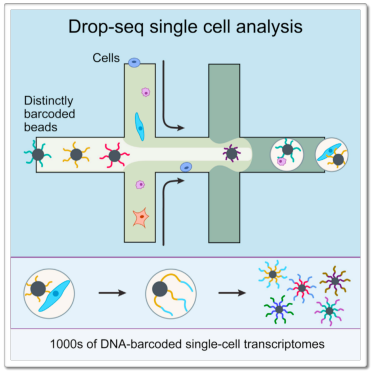


- **Smart-seq（Switching mechanism at 5' end of the RNA transcript）的优点：**
  1. **使用低至50 pg的起始材料。**
  2. **不需要知道mRNA的序列。**
  3. **不再需要纯化步骤。**
  4. **转录本的覆盖度改善。**
  5. **高水平的可定位序列。**
- **缺点：**
  1. **并非链特异的。**
  2. **只测序poly(A)+ RNA。**

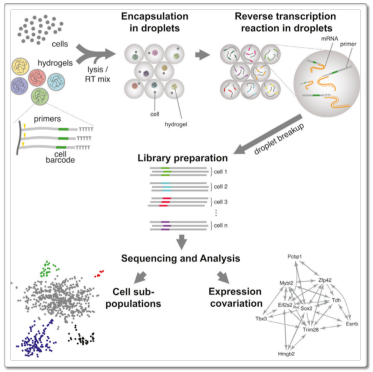
单细胞RNA-seq技术举例—Drop-seq/inDrop



单细胞RNA-seq技术举例—Drop-seq/inDrop

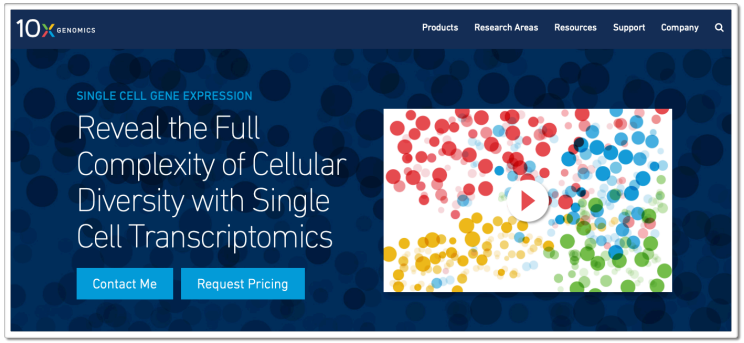


Drop-seq

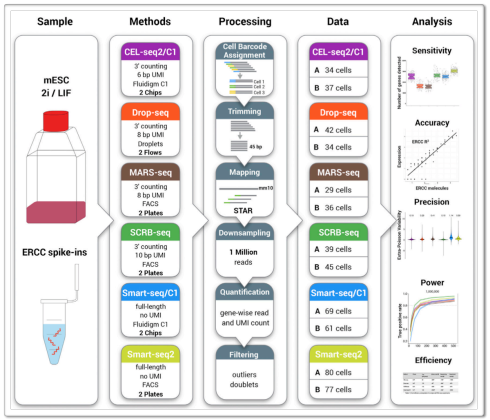


inDrop

单细胞RNA-seq技术举例—10x genomics



单细胞RNA-seq技术比较

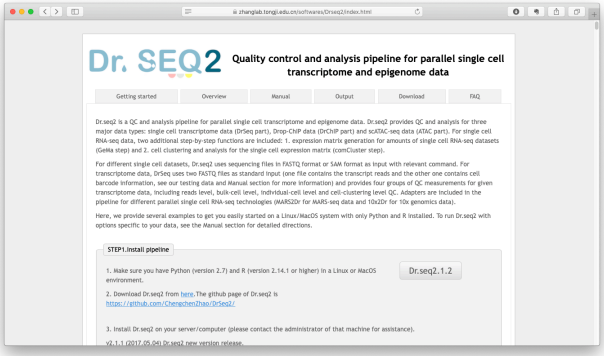


- 所有方法的准确性相似；
- Smart-seq2的灵敏度最高；
- CEL-seq2最为精确；
- Drop-seq最为高效。

常见针对单细胞 RNA-seq 的分析工具

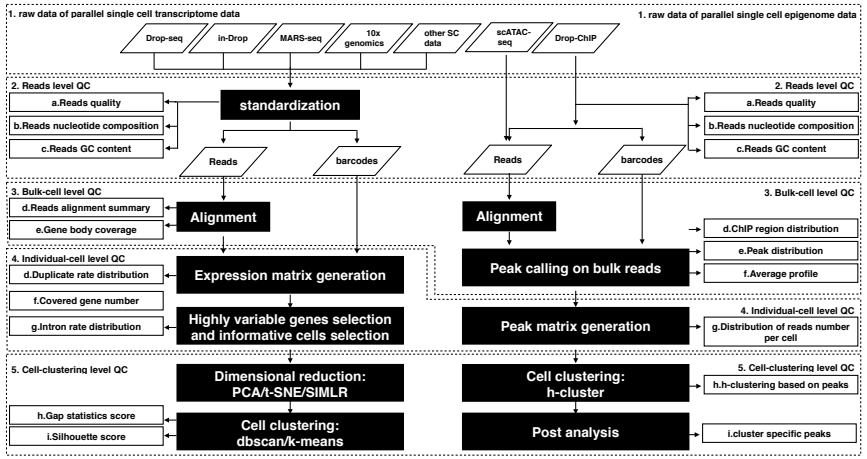
Name	Multiple data type supporting	Supporting parrell single cell data (e.g. Drop-seq, Mars-seq)	Supporting single cell epigenome data (e.g. scATAC and Drop-ChIP)	Reads level QC	Individual cell level QC	Highly variable gene detection	Noise reduction	Informative cell selection	Cell clustering (sub cell type identification)	Differential expressed gene detection	Pseudotemporal ordering
Dr.seq2	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔	
Dr.seq		✔		✔	✔	✔	✔	✔	✔	✔	
BASiCS						✔	✔		✔		
scLVM						✔	✔			✔	
SINCERA				✔	✔	✔	✔		✔	✔	
OEFinder						✔	✔				
ZIFA							✔		✔		
Destiny							✔		✔		
SNN-Cliq							✔		✔		
RaceID							✔		✔	✔	
SCUBA							✔		✔	✔	✔
BackSPIN									✔	✔	
PAGODA						✔	✔		✔		
MAST										✔	
SCDE									✔	✔	
scDD										✔	
Monocle											✔
Waterfall									✔		✔
Sincell									✔		✔
Oscope									✔		✔
Wanderlust									✔		✔
CellTree									✔		✔
SinQC				✔	✔		✔				
ASAP							✔		✔	✔	

Dr.seq2

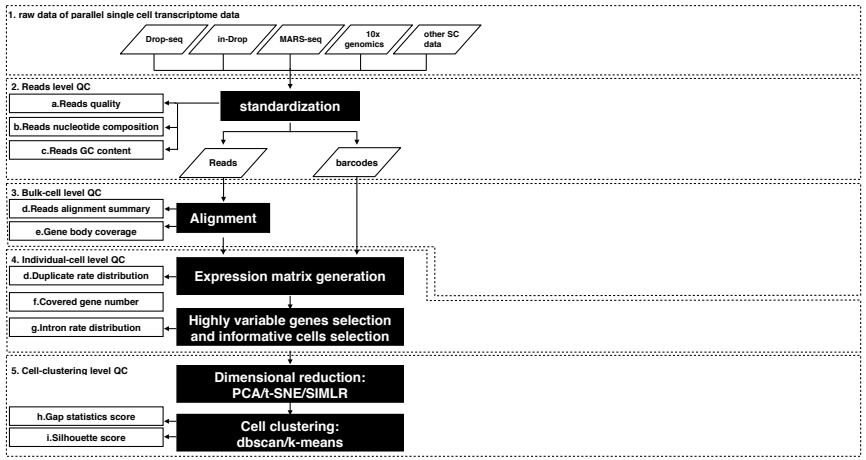


- 1.Dr.seq2支持不同类型的单细胞转录组学数据以及单细胞表观遗传组学数据；
- 2.Dr.seq2提供了详尽的质量控制报告以及细胞聚类分析结果。

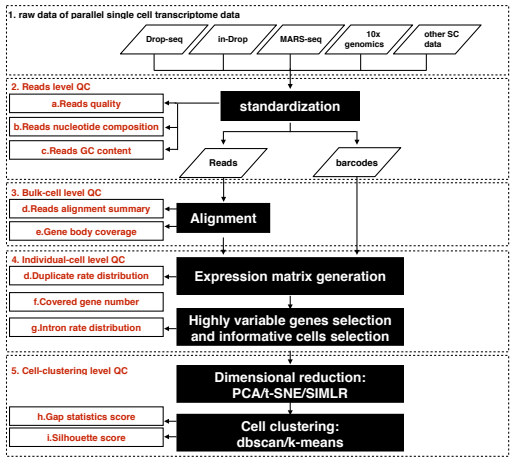
Dr.seq2质量控制及分析流程



Dr.seq2质量控制及分析流程



Dr.seq2质量控制及分析流程



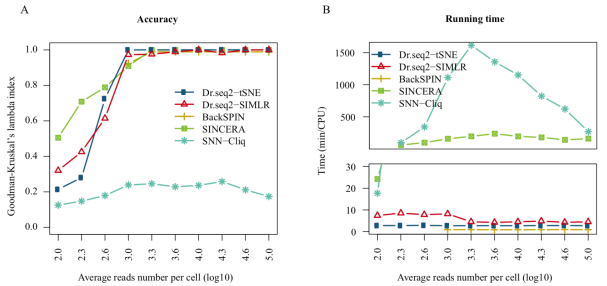
（1）原始测序片段层面。对二代测序数据的质量进行整体的评估；

（2）RNA文库建立层面。将测序数据进行序列回帖到特定基因组后，可以得到每个基因的表达值，此时通过一些指标进行细胞质量好坏的判断；

（3）单个细胞测序质量层面；

（4）细胞聚类层面。

Dr.seq2中集成的聚类功能比较



Dr.seq2的下载与安装

- 安装环境与操作系统
  - Linux or MacOS
  - Python (version 2.7) and R (version 2.14.1+)
- 下载Dr.seq2
- 安装Dr.seq2
- 准备必要组件及参考基因组文件

Dr.seq2的下载与安装

- 安装环境与操作系统
- 下载Dr.seq2

```
$ apt-get install wget  #Skip this step,if you can use wget
$ wget https://zhanglab.tongji.edu.cn/softwares/Drseq2/download/Dr.seq2.1.2.zip
$ unzip Dr.seq2.1.2.zip
$ cd Dr.seq2.1.2
```

- 安装Dr.seq2
- 准备必要组件及参考基因组文件

Dr.seq2的下载与安装

- 安装环境与操作系统
- 下载Dr.seq2
- 安装Dr.seq2

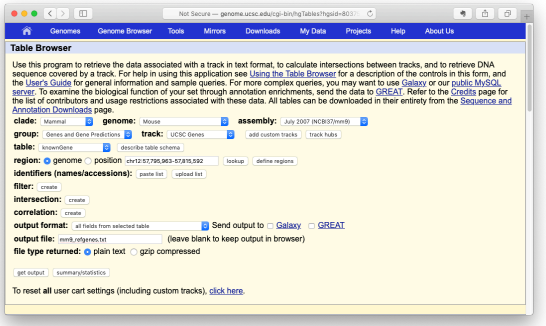
```
# For root users:
$ sudo python setup.py install
# For  other users:
$ python setup.py install --prefix /home/DrSeq2
# here you can replace "/home/DrSeq2" with any location you want
$ export PATH=/home/DrSeq2/bin:$PATH
# setup PATH, so that system knows where to find executable files
$ export PYTHONPATH=/home/DrSeq2/lib/python2.7/site-packages:$PYTHONPATH
# setup PYTHONPATH, so that Dr.seq2 knows where to import modules
# Test:
$ DrSeq —help
```

- 准备必要组件及参考基因组文件

Dr.seq2的下载与安装

- 安装环境与操作系统
- 下载Dr.seq2
- 安装Dr.seq2
- 准备必要组件及参考基因组文件
  - Prepare the mapping software
    - We can use bowtie2, hisat2 or STAR as maptool.
  - Prepare mapping software index
    - different mapping softwares should use their own index.
  - Obtain gene annotation information
    - Gene annotation information for the species should be prepared.

准备基因组注释文件



bin	name	chrom	strand	txStart	txEnd	cdsStart	cdsEnd	exonCount	exonStarts	exonEnds	score	name2	cdsStartStat	cdsEndStat	exonFrames
0	NM_001282945	chr1	-	134199214	134235227	134202950	134234355	3	134199214,134234014,134235227,	134203590,134234446,134235457,	0	Adora1	cmpl	cmpl	2,0,-1,

mm9_refgenes.txt

运行Dr.seq2

- 运行 Dr.seq2（针对转录组数据的命令为 DrSeq）
  - simple mode that can run Dr.seq2 quickly
  - standard mode that can run Dr.seq2 conveniently and flexibly

DrSeq simple模式

```
$ DrSeq simple -h
usage: DrSeq simple [-h] -b BARCODE -r READS -n NAME
                    [--cellbarcoderange CBR]
                    [--umirange UMIR] [-g GA]
                    [--maptool {STAR,bowtie2,HISAT2}]
                    [--checkmem {0,1}]
                    [--mapindex MAPINDEX]
                    [--thread P] [-f] [--clean]
                    [--select_cell_measure {1,2}]
                    [--remove_low_dup_cell {0,1}]
```

DrSeq simple模式下的必要参数

- -b BARCODE, --barcode BARCODE
  - barcode fastq file before any filtering step, only accept .fastq format
- -r READS, --reads READS
  - reads fastq file, accept raw fastq input or aligned sam format,format(fastq,sam) fixed by extension(.fastq or .sam)
- -n NAME, --name NAME
  - name of you config file and output dir, name only , no extension. The output files will be named like name.pdf, name.txt …
- -g GA, --gene_annotation GA
  - gene annotation file, the annotation file can be download from UCSC
- --maptool {STAR, bowtie2, HISAT2}
  - choose mapping software for alignment, default is HISAT2, you can also choose bowtie2 or STAR as another option.
- --mapindex MAPINDEX
  - mapping index folder, there should be a mm10.star folder under mapindex folder if you use STAR to map to mm10 genome, mm10.bowtie2 folder if use bowtie2.

DrSeq simple运行举例

```
$DrSeq simple -b /home/user/drseq_test_1.fastq
    -r /home/user/drseq_test_2.fastq
    -n simple_mode_test
    -g /home/user/annotation/mm9_refgenes.txt
    --maptool bowtie2
    --mapindex /home/user/bowtie2_index/mm9
```


DrSeq simple模式下的可选参数

- -h, --help
  - show help message and exit
- --cellbarcoderange CBR
  - specify the length of your cell barcode , default is 1:12, 12(cellbarcode) + 8(umi) = 20 (barcodefastq)
- --umirange UMIR
  - specify the length of your UMI , default is 13:20, 12(cellbarcode) + 8(umi) = 20 (barcodefastq)
- --checkmem {0,1}
  - (default is 1 (on), only take effect when maptool = STAR) Dr.seq will check your total memory (if turned on, set 1, default) to make sure its greater 40G if you choose STAR as mapping tool.
- --thread P
  - number of alignment threads to launch, ignored for sam input
- -f, --force_overwrite
  - specify the config file to create output folder , this cmd will rm existing result if set True ~!!
- --clean
  - remove intermediate result generated during Dr.seq,default is No
- --select_cell_measure {1,2}
  - Method to select STAMPs from cell_barcodes, choose from 1 or 2 (default is 1), 1: Cell_barcodes with more than 1000 genes covered are selected. 2: Top 1000 cell_barcodes with highest umi count will be selected
- --remove_low_dup_cell {0,1}
  - discard cell barcodes with low duplicate rate (< 0.1), set 1 to turn on this function, set 0 (default) to turn off. May not effective for samples with low sequencing depth

DrSeq standard模式

```
DrSeq [-h] [-v] {gen,run} ...
    gen         generate a template of config file
    run         run pipeline with a config file input
```

使用DrSeq gen模块生成参数文件

```
DrSeq gen [-h] -n CONFIG_NAME

arguments:
  -h, --help            show this help message and exit
  -n CONFIG_NAME, --name CONFIG_NAME
                        name of your config file : config_name.conf
```


使用DrSeq run模块运行

```
DrSeq run [-h] -c CONFIG [-f] [--clean]

optional arguments:
  -h, --help      show this help message and exit
  -c CONFIG, --config CONFIG
                  specify the config file, -c config_name.conf
  -f, --force_overwrite
                  The pipeline will over write output result if the
                  output folder is already exist
  --clean         remove intermediate result generated during
                  Dr.seq,default is NO
```


利用standard模式运行 Dr.seq2的例子

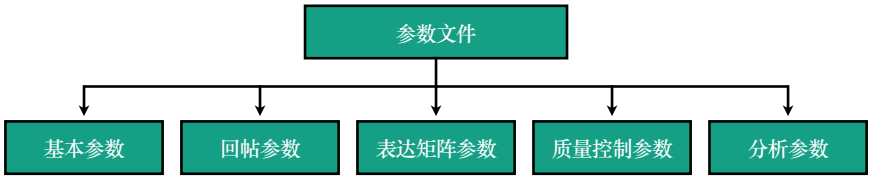
- 首先创建参数模板文件。

```
$ DrSeq gen -n my_config.conf
```

- 然后按照需要修改 my_config.conf 中的参数，之后运行 Dr.seq2。

```
$ DrSeq run -c my_config.conf
```


参数文件中的参数说明



基本参数

```
[General]
barcode_file =
cell_barcode_range = 1:12
umi_range = 13:20
reads_file =
outputdirectory =
outname =
gene_annotation = /home/user/annotation/mm10_refgenes.txt
png_for_dot = 0
```

- barcode_file =
  - # fastq file only, by default, every barcode in fastq file consist of 20 bp including 12bp cell barcode(1-12bp) and 8bp UMI(13-20bp).

基本参数

```
[General]
barcode_file =
cell_barcode_range = 1:12
umi_range = 13:20
reads_file =
outputdirectory =
outname =
gene_annotation = /home/user/annotation/mm10_refgenes.txt
png_for_dot = 0
```

- cell_barcode_range = 1:12
- umi_range = 13:20
  - # Location of cell barcode and UMI,. by default they are 1:12 and 13:20.
  - For example:
  - If the barcode in your fastq is AAAAAAAAAAAACCCCCCCC,then the cell_barcode of this reads is A 1:12 and the UMI is C 13:20

基本参数

```
[General]
barcode_file =
cell_barcode_range = 1:12
umi_range = 13:20
reads_file =
outputdirectory =
outname =
gene_annotation = /home/user/annotation/mm10_refgenes.txt
png_for_dot = 0
```

- reads_file =
  - # Accept raw sequencing file (fastq) or aligned file(sam), file type is specified by extension.
- outputdirectory =
  - # Directory for all result. default is current dir "." if user left it blank.
- outname =
  - # Name of all your output results, your results will looks like outname.pdf, outname.txt

基本参数

```
[General]
barcode_file =
cell_barcode_range = 1:12
umi_range = 13:20
reads_file =
outputdirectory =
outname =
gene_annotation = /home/user/annotation/mm10_refgenes.txt
png_for_dot = 0
```

- gene_annotation =
  - # Gene model in full text format, download from UCSC genome browser.
- png_for_dot = 0
  - #Only works for dotplots in individual cell QC. set 1 to plot dotplots in individual cell QC in png format, otherwise they will be plotted in pdf format.

回帖参数

```
[Step1_Mapping]
mapping_software_main = hisat2
checkmem = 1
mapping_p = 8
mapindex = /home/user/hisat2_index
q30filter = 1
```

- mapping_software_main = hisat2
  - # name of your mapping software, choose from hisat2, STAR and bowtie2
- checkmem = 1
  - # only take effect when mapping_software_main = STAR.Dr.seq2 will check your total memory to make sure its greater 40G if you choose STAR as mapping tool.

回帖参数


```
[Step1_Mapping]
mapping_software_main = hisat2
checkmem = 1
mapping_p = 8
mapindex = /home/user/hisat2_index
q30filter = 1
```


- mapping_p = 8
  - # Number of alignment threads to launch alignment software
- mapindex = /home/user/hisat2_index
  - # Mapping index of your alignment tool
- q30filter = 1
  - # Use q30 criteria (Phred quality scores) to filter reads in sam file, set this parameter to 1(default) to turn on this option, set 0 to turn off. Default is 1

表达矩阵参数

```
[Step2_ExpMat]
filterttsdistance = 0
ttsdistance = 400
covergncutoff = 100
duplicate_measure = 2
umidis1 = 0
```

- filterttsdistance = 0
  - #filterttsdistance: Whether discard reads far away from tts, choose from 0 or 1, set 1 to turn on this function, default is 0
- ttsdistance = 400
  - #Default filter distance is 400bp, only take effect when filterttsdistance = 1 is set

表达矩阵参数


```
[Step2_ExpMat]
filterttsdistance = 0
ttsdistance = 400
covergncutoff = 100
duplicate_measure = 2
umidis1 = 0
```


- covergncutoff = 100
  - #Only plot cell barcodes with more than 100 (default) genes covered in the dotplots in individual-cell QC step.

表达矩阵参数

```
[Step2_ExpMat]
filterttsdistance = 0
ttsdistance = 400
covergncutoff = 100
duplicate_measure = 2
umidis1 = 0
```

- duplicate_measure = 2
  - # method to consider duplicate reads in each cell barcodes when generate expression matrix,
  - 1 : combine duplicate reads with same UMI and same genomic location (position and strand)
  - 2(default) : only consider UMI (combine duplicate reads with same UMI)
  - 3 : only consider genomic location only (combine duplicate reads with same genomic location)
  - 0 : do not remove(combine) duplicate reads (keep all duplicate reads, skip combination step)

表达矩阵参数

```
[Step2_ExpMat]
filterttsdistance = 0
ttsdistance = 400
covergncutoff = 100
duplicate_measure = 2
umidis1 = 0
```

- umidis1 = 0
  - #only take effect when duplicate_measure = 1 or 2, ignore this parameter if duplicate_measure = 0 or 3
  - Set 1 :if two reads from same cell barcode have same genomic location(chrom,position,strand) and their umi distance <= 1, regard them as duplicate reads and combine them.
  - Set 0 (default) : regard them as duplicate reads only when their umi distance = 0 and have genomic location.

质量控制参数

```
[Step3_QC]
select_cell_measure = 1
covergncluster = 1000
topumicellnumber = 1000
remove_low_dup_cell = 0
non_dup_cutoff = 0.1
```

- select_cell_measure = 1
  - # Method to select real cells from cell_barcodes, choose from 1/2
  - 1 (default): Cell_barcodes with more than 1000 genes covered are selected as real cells for following analysis including dimentional reduction and clustering, cutoff of covered gene number(1000, default) is determined in parameter "covergncluster"
  - 2 : Top 1000 cell_barcodes with highest umi count will be selected as real cells, number of highest umi cell(1000, default) is determined in parameter "topumicellnumber". Suitable for Drop-seq sample with known cell number

质量控制参数

```
[Step3_QC]
select_cell_measure = 1
covergncluster = 1000
topumicellnumber = 1000
remove_low_dup_cell = 0
non_dup_cutoff = 0.1
```

- covergncluster = 1000
  - # covergncluster: cell_barcodes with more than 1000 (default) genes covered will be selected as STAMP barcodes for clustering
  - This parameter takes effect only when "select_cell_measure" is set to 1
- topumicellnumber = 1000
  - # topumicellnumber: Top 1000(default) cell_barcodes with highest umi count will be selected as STAMP barcodes for clustering
  - This parameter takes effect only when "select_cell_measure" is set to 2

质量控制参数

```
[Step3_QC]
select_cell_measure = 1
covergncluster = 1000
topumicellnumber = 1000
remove_low_dup_cell = 0
non_dup_cutoff = 0.1
```

- remove_low_dup_cell = 0
  - # A group of cell barcodes have almost no duplicate reads, which show clearly different pattern from STAMP barcodes, we set an option to remove these cell_barcodes for analysis step
- non_dup_cutoff = 0.1
  - # Only take effect when remove_low_dup_cell is turned on. Cutoff of low-duplicate cell_barcode (default is 0.1), a cell_barcode will be defined as low-duplicate cell_barcode if its duplicate rate < 0.1.

分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- dimensionreduction_method = 1
  - #Method for dimension reduction.chosse from 1(default),2,3
    - 1(default) : t-SNE
    - 2 : SIMLR
    - 3 : PCA

分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- highvarz = 1.64
  - #Cutoff for high variance selection. in the first step we divide all genes to 20 groups based on average expression level across all individual cells. Then in each group we select genes whose z-normalized CV(var/mean) >= 1.64
- selectpccumvar = 0.5
  - #Select topN PC until they explain 50% (default is 0.5, eg. 0.3 for 30% variance) of total variance.

分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- pctable = 1
  - # choose from 0(default) and 1, you can turn on this function(set 1) to output a 2 column table for PC1 v.s PC2
    - This paramter is designed for users who prefer and conduct following analysis based on PCA output
    - 0(default) : Do not output PCA result, only output PCA + 2 dimentional t-SNE result
    - 1 : Output PC1 v.s. PC2 table in addition to t-SNE result.

分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- cortable = 1
  - #Choose from 0(default) and 1, you can turn on this function(set 1) to output a cell to cell correlation matrix.
    - Correlation is based on log scale TPM(trsncript per million reads).
    - User can turn on this function to generate cell to cell correlation matrix for custom analysis

分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- clustering_method = 1
  - #Method for cluster cells based on t-SNE output
    - Choose from 1(default) ,2 ,3 and 4.
    - 1(default) : k-means, use Gap statistics followed by our "first stable Gap" method to determine k
    - 2 : k-means, but use Gap statistics followed by tridiitional "Tibs2001SEmax" method to determine k
    - 3 : k-means with custom determined k, k value is defined in following parameter "custom_k", this option is designed for users who know the number of subgroup of the drop-seq sample
    - 4 : (make sure your R enviroment has library "fpc" installed) Use dbscan as clustering method, the (eps) parameter is defined in following parameter "custom_d"

分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- maxknum = 100
  - # Maximum k number for gap statistics, only take effect when clustering_method = 1 or 2
- custom_k = 5
  - # Only take effect when clustering_method = 3, cells will be clustered to N group based on t-SNE result according to user determined k
- custom_d = 2
  - # Only take effect when clustering_method = 4, refer to the parameter "eps"of dbscan. By default we set it to 2 according to the orginal Drop-seq paper, but it varies a lot between differen datasets.

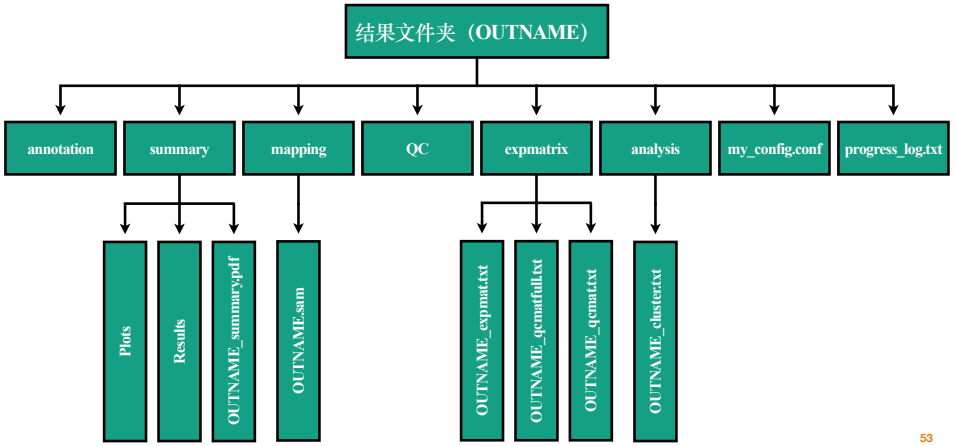
分析参数

```
[Step4_Analysis]

dimensionreduction_method = 1
highvarz = 1.64
selectpccumvar = 0.5
pctable = 1
cortable = 1
clustering_method = 1
maxknum = 100
custom_k = 5
custom_d = 2
rdnumber = 1007
seed = 199
```

- rdnumber = 1007
  - #Set random number times
- seed = 199
  - #Set initial seed to keep your result reproducible

Dr.seq2输出结果的结构



Dr.seq2输出结果的结构

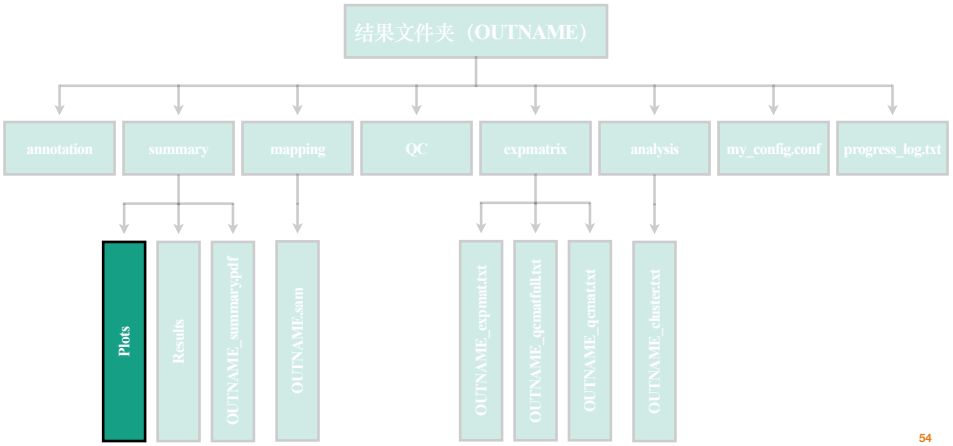
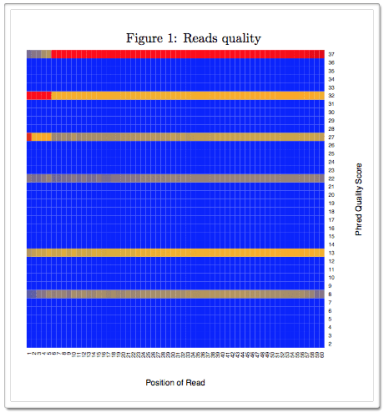
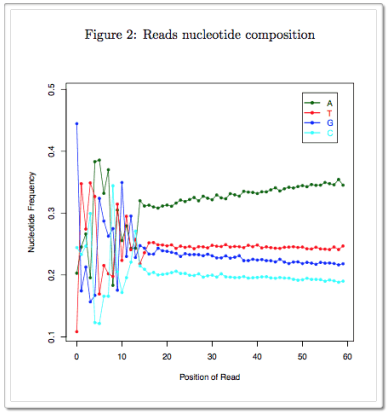


Figure1_quality_heatmap.pdf



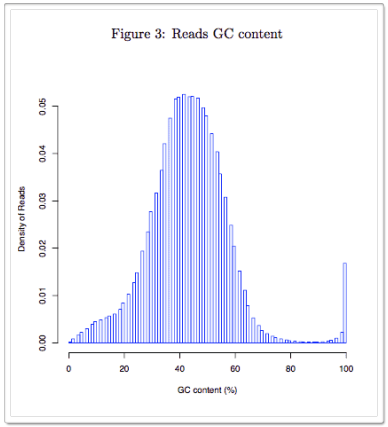
- Reads quality is one of the basic reads level quality control methods.
- Color in the heatmap represented frequency of this quality score observed at this position.
- Red represented higher frequency while blue was lower frequency.

Figure2_NVC.pdf



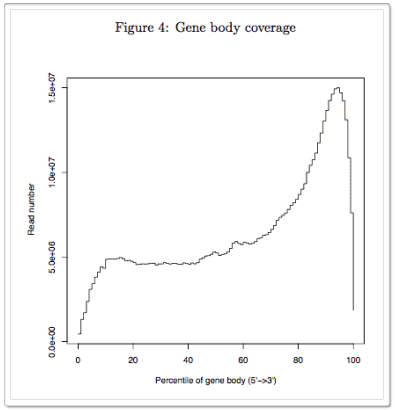
- We assessed the nucleotide composition bias of a sample.
- The proportion of four different nucleotides was calculated at each position of reads.
- Theoretically four nucleotides had similar proportion at each position of reads.
- You may observe higher A/T count at 3’end of reads because of the 3’end polyA tail generated in sequencing cDNA libaray, otherwise the A/T count should be closer to C/G count.

Figure3_GC.pdf



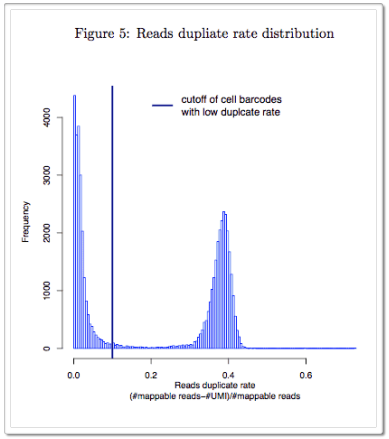
- Distribution of GC content of each read. This module measures the general quality of the library.
- If the distribution looks different from a single bell(too sharp or too broad) then there may be a problem with the library.
- Sharp peaks on an otherwise smooth distribution are normally the result of a specific contaminant (adapter dimers for example), which may well be picked up by the overrepresented sequences module.
- Broader peaks may represent contamination with a different species.

Figure4_GBcover.pdf



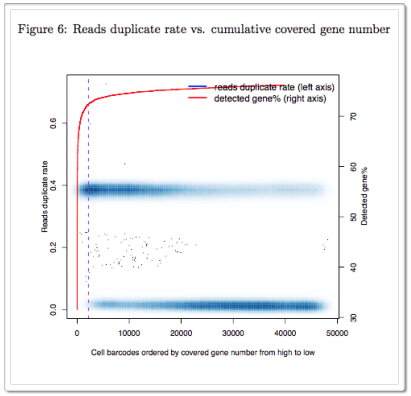
- Aggregate plot of reads coverage on all genes. This module measures the general quality of Drop-seq data.
- Theoretically we observe a unimodal(single bell) distribution, but for Drop-seq sample an enrichment at 3'end is observed due to library preparation using oligo-dT primers.
- In any case you should observed a smooth distritbuion. If loss of reads or spike were observed in certain part of gene body (e.g. middle or 3'end of gene body), poor quality of your library was indicated.

Figure5_duprate.pdf



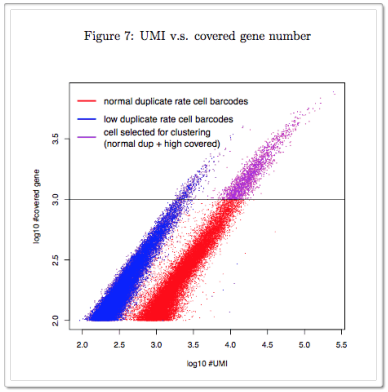
- Drop-seq technology has an innate advantage of detect duplicate reads and amplification bias because of the barcode and UMI information.
- This module displays the distribution of duplicate rate in each cell barcode and helps to discard cell barcodes with low duplicate rate.
- We plotted the distribution of duplicate rate in each cell barcode

Figure6_cumUMI_duprate.pdf



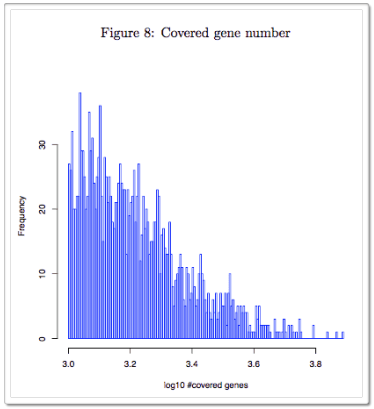
- Reads duplicate rate versus cumulative covered gene numbers.
- This module measures whether each of your single cell was sequenced and was clearly separated from empty cell barcodes.
- Cell barcodes were ranked by the number of covered genes.
- The duplicate rate (y-axis, left side) was plotted as a function of ranked cell barcode. Red curve represented the number of genes covered by top N cell barcodes (y-axis, right side). N was showed by x-axis.
- Theoretically you will observed a "knee" on your cumulative curve (slope = 1 on the curve) and the cutoff of your selected STAMPs (dash line) should be close to the "knee".

Figure7_umi_coverGN.pdf



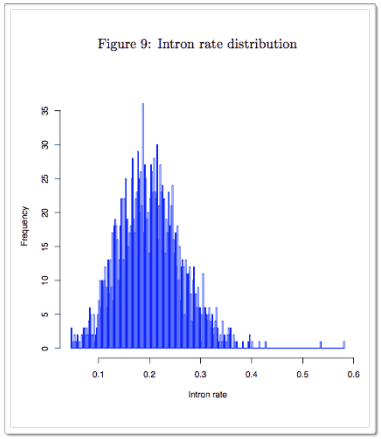
- Covered gene number was plotted as a function of the number of UMI (i.e. unique read).
- This module measures the quality of Drop-seq experiment and helps to distinguish STAMPs from empty cell barcodes.
- We observed a clearly different pattern for two groups of cell barcodes with different reads duplicate rate (blue dots versus red and purple dots). Purple dots represented the selected STAMPs for the cell-clustering analysis. By default we select STAMPs with 1000 gene covered after discarding low duplicate cell barcodes.

Figure8_coverGN.pdf



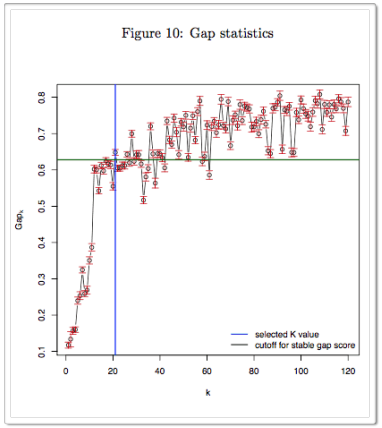
- Histogram of covered gene number of selected STAMPs.
- The module tell you whether the selected STAMPs have sufficient reads coverage. By default Dr.seq select STAMPs with cell barcodes with >= 1000 genes covered.
- If you select STAMPs with highest reads count (“select cell measure” = 2), then you should check this figure to make sure the STAMPs you select have enough gene covered

Figure9_intronrate.pdf



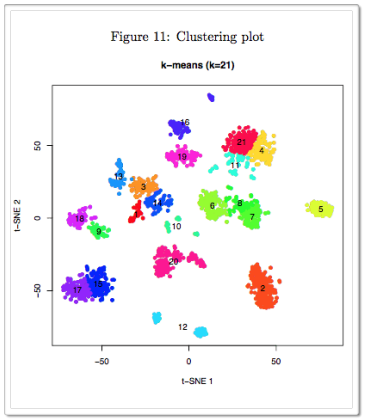
- Intron rate is a effective method to measure the quality of a RNA-seq sample.
- We plotted a histogram of intron rate of every STAMP barcodes to check whether reads from each STAMPs enriched in the exon region.
- High intron rate (e.g. >= 30%) indicates low quality of cDNA in each STAMPs (the low quality could be caused by different problem, for example contaminant).
- We suggest redo the Drop-seq experiment if most of selected STAMPs have high intron rate and low covered gene number.

Figure10_GapStat.pdf



- We conducted a k-means clustering based on t-SNE dimensional reduction output to measure sample’s ability to be separated to different cell subtypes.
- Gap statistics was performed to determine the best k in k-means clustering.
- In general, decreasing pattern (usually k <= 2) is observed for pure cell type or cell line data, while increasing pattern with bigger k should be observed for mix cell types (or cell subtypes) data.
- If the cluster number predicted from the Gap statistics has great difference from what you expect, it indicated that your cells were not well characterized and separated by the Drop-seq experiment (because of the contaminant or the low capture efficiency of Droplets).

Figure11_cluster.pdf



- Scatter plot represented visualization of t-SNE dimensional reduction output of selected STAMP barcodes.
- STAMP barcodes were colored according to the clustering result and cluster numbers were printed in the center of each cluster.
- This figure is mainly for visualization and help you to know how their data looked like. If you want to combine some small groups which are close to each other, you can use the cluster matrix (named “cluster.txt”) Dr.seq provided to conduct your own analysis.